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EDUCATION

- University of Pittsburgh (Swanson School of Engineering/Frederick Honors Col.)** **Pittsburgh, PA**
B.S. - Computer Engineering (Autonomous Systems Focus); GPA: 3.99/4.00 *August 2022 - April 2026*
- Spring-Ford Area High School** **Royersford, PA**
High School Degree (Summa Cum Laude); GPA: 100.88/100.00 *August 2018 - June 2022*

PROFESSIONAL EXPERIENCE

- Carnegie Mellon University Robotics Institute** **Pittsburgh, PA**
Undergraduate Roboticist Researcher *April 2023 - Present*
 - ZOË 2 Rover:** (tinyurl.com/zoe2rover)
 - Assisting in the development of software architecture and the robotic controller implementation for a novel 4-wheeled rover with a unique passive steering mechanism.
 - Enabling low-level software development for the rover using ROS2 and C++ to interface with motor controllers and encoders via CANOpen protocol.
 - Conducted motor and motor controller datasheet specifications validation via physical testing.
 - Underwater Snake Robot:** (tinyurl.com/humrsCMU)
 - Implemented High-Frequency Injection methods in Brushless DC thrusters to achieve control at low/zero speeds thus reducing minimum speed by 80% allowing for reduced oscillations in station-keeping control loops.
 - Deployed station-keeping feature using AprilTags, IMU readings, and Nested PID Controllers to perform robot state-estimation underwater.
 - Conducted major repairs on the buoyancy module of the robot and assisted in continual electrical/hardware maintenance.
 - Apple's E-Waste Recycling Project:** (tinyurl.com/applecmu)
 - Labeled datasets consisting of 4000+ images for Machine Learning Models to detect screws in e-waste images.
 - Integrated ROS1 and Python packages to track AprilTags using a Realsense camera for localization of UR3 robotic arm.
 - Manufactured custom AprilTags using lasercutters and sheet metal manufacturing methods.
- ESTAT Actuation** **Pittsburgh, PA**
Robotics Developer Co-op *May 2025 - Present*
 - Building and commissioning multiple robotic test stands expanding clutch testing capacity by 133%, covering mechanical assembly, electrical wiring, programming, and documentation for long-term use and potential sales to external customers.
 - Directing the assembly and delivery of 6 custom battery boxes for the Suzuki Moqba robot at the Japan Mobility Show. Overseeing quality control, leading debugging efforts to resolve heating issues, and ensuring a professional final build.
 - Designing and assembling customer facing demos for prototype versions of encoder and clutch integrated products.
 - Testing new generation rotary clutch prototypes to improve reliability and evaluate longevity. Creating documentation explaining testing procedures, and using these prototypes to inform design decisions for next generation production clutches.

LEADERSHIP & ACTIVITIES

- Robotics and Automation Society (University Rover Challenge)** **Pittsburgh, PA**
Chief Engineer/Integration Lead *August 2022 - Present*
 - Leading a multidisciplinary team of 30 students in engineering a rover, coordinating integration efforts across mechanical, electrical, software, and bioscience teams. (tinyurl.com/roverimages)
 - Mentoring younger students in Onshape, KiCAD, and general manufacturing processes through hands-on building.
 - Securing and managing over \$7,000 in funding and sponsorships to support team development and future growth.
 - Proposed and spearheaded the development of a prototype robotic hand utilizing pneumatics. (tinyurl.com/hydraarm)
- Formula SAE - Panther Racing** **Pittsburgh, PA**
R&D Engineer *August 2022 - March 2024*
 - Developed a 3D-printed e-brake bias system for a Formula SAE racecar using Solidworks, improving performance and usability; won the FSAE Innovation Award for implementation of the project. (tinyurl.com/ebrakeb)
 - Worked to design and code prototypes of slip-angle sensors which would allow for validation of slip angle using Raspberry Pi, mouse sensors, digital cameras, and IMUs.
- FIRST Robotics** **Exton, PA**
Team Captain/Design Lead *January 2020 - August 2022*
 - Led a team of 40 students and qualified for the Worlds level of competition, the highest win percentage since 2005, and a top 5% ranking globally.
 - Utilized Solidworks to design and develop competition and award-winning robots.
 - Designed and manufactured experimental west-coast drive chassis using Solidworks.

- Led a small group of students in developing a website for an offseason competition to track climate conditions. Placed 2nd and won tablets for the team which I donated to the team.
- Competed in weekend long CADathon designing a robot for moving weights and launching footballs. Placed 17th out of a 100 teams. (youtu.be/XIL8EZkmz6o?t=2461)

Spring-Ford Technology Club (VEX Robotics)

Royersford, PA

August 2018 - June 2022

- *Vice President/Team Captain*
 - Organized VEX robotics competition event for 60+ teams and received a scholarship for outstanding leadership within the club. (www.youtube.com/watch?v=iPC_ZToDl7I)
 - Designed, built, and coded 3 robots during my time, qualifying for the state level of competition each year.
 - Maintained an engineering notebook to track engineering decisions and progress through each season.
 - Mentored freshmen students on robot design and manufacturing through workshops and hands-on building.

PROJECTS

- **Micromouse Robot:** Designed, assembled, and programmed a custom Micromouse robot through iterative PCB development, integrating motor control, IMU, IR sensors, and a display system with low-level embedded programming. Implemented PID control loops for navigation. Placed 3rd in competition at Stony Brook University IEEE conference. (tinyurl.com/goyalmm)
- **555 Timer Roulette Board:** Designed a PCB LED Roulette Wheel using 555 timer and CD4017 counter, featuring dynamic speed control using capacitive timing and a 3D-printed enclosure. (tinyurl.com/555roulette)
- **VoltVitals Embedded Board:** Designed and prototyped an ESP32 based circuit with web interface to measure home outlet voltage and report over Wi-Fi, collaborating with FirstEnergy engineers to enable early outage detection. (tinyurl.com/voltvitals)
- **An Artistic take on Bipedal Robots:** Created a bipedal robot using Onshape and Arduino to carry a symbolic lantern given to all freshmen, reimagining a campus tradition through an artistic and technological lens. (tinyurl.com/gbipedal)
- **STM-32 Elevator Simulator:** Designed and implemented software architecture in ARM Assembly to operate a physical elevator simulator PCB. (tinyurl.com/stmelevator)
- **Custom Cane:** Designed and fabricated a walker-cane fusion to increase bathrooms accessibility for wheelchair users. Won first place at the Senior Design Expo. (tinyurl.com/CustomCane)
- **Silicon Prosthetic Hand (High School Independent Research, Solidworks):** Designed and manufactured prototype prosthetic hand with silicone soft actuators and tested with human participants for an AP Research Project. (Link: <https://tinyurl.com/myprosthetic>)

PUBLICATIONS

- M. Nye, A. Raji, A. Saba, E. Erlich, R. Exley, **A. Goyal**, A. Matros, R. Misra, M. Sivaprakasam, M. Bertogna, D. Ramanan, and S. Scherer, “**BETTY Dataset: A Multi-modal Dataset for Full-Stack Autonomy**,” in Proc. IEEE Int. Conf. Robot. Autom. (ICRA), 2025. [Online]. Available: (arxiv.org/abs/2505.07266)

SKILLS AND AWARDS

- **Software:** Linux, ROS1, ROS2, Github, Python, C++, C, MATLAB, ARM and RISC-V Assembly, CANOpen, Google Colab
- **Electrical:** Altium, KiCAD, LTSpice, Soldering, Arduino, ESP32
- **Mechanical:** Solidworks, Onshape, Milling, Lathe, MIG Welding, Laser Cutting, General Shop Tools
- **Awards:** Micromouse Competition 3rd place, SSOE Design Expo 1st place Art of Making, FSAE Innovation Award, Eagle Scout

VOLUNTEER EXPERIENCE

- **American Red Cross** **Royersford, PA**
Blood Donor Ambassador *Jan 2019 - Present*
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- **Boy Scouts of America** **Royersford, PA**
Eagle Scout Project *Jan 2018 - Present*
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